

ME23312 ENGINEERING THERMODYNAMICS

Unit II – Second Law and Availability Analysis · Presentation 5 of 10

The Clausius Inequality and Entropy

$\oint \delta Q/T$ for reversible and irreversible cycles · the Clausius inequality · entropy is a property · $dS = (\delta Q/T)_{\text{rev}}$ · the T–s diagram · heat as an area

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2. $\oint \delta Q/T$ for a reversible engine
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The Clausius inequality

From two reservoirs to any cycle

- The Carnot relations concern heat exchanged with **two** reservoirs at fixed temperatures
- Clausius generalised them to **any** cycle exchanging heat at **any** number of temperatures
- The result – the Clausius inequality – is the doorway to the property **entropy**
- Sign convention: heat received positive, heat rejected negative

Reversible engine between T_1 and T_2

$$Q_1/T_1 = Q_2/T_2 \Rightarrow Q_1/T_1 + (-Q_2)/T_2 = 0$$

- From the Kelvin scale, $(Q_1/Q_2)_{\text{rev}} = T_1/T_2$
- With the sign convention the rejected heat is $-Q_2$, so the algebraic sum of Q/T around the cycle is **zero**

Irreversible engine between T_1 and T_2

$$\eta_I < \eta_R \Rightarrow Q_2/Q_1 > T_2/T_1 \Rightarrow Q_1/T_1 + (-Q_2)/T_2 < 0$$

- Carnot theorem: $1 - Q_2/Q_1 < 1 - T_2/T_1$
- The irreversible engine rejects **more** heat per unit heat received, so $Q_2/T_2 > Q_1/T_1$
- The algebraic sum of Q/T around the cycle is **negative**

An arbitrary cycle

- Heat received and rejected at continuously varying temperatures
- Divide the cycle by many closely spaced reversible adiabatics into elementary strips
- Each strip, closed by two short reversible isotherms, is an **elementary Carnot cycle** taking δQ_1 at T_1 and rejecting δQ_2 at T_2
- Isotherm segments of neighbouring strips cancel in pairs; sum over all strips and pass to the limit

The Clausius inequality

$$\oint \delta Q/T \leq 0$$

= 0 reversible cycle, < 0 irreversible cycle

- T is the absolute temperature where δQ crosses the boundary
- $\oint \delta Q/T$ can **never be positive** – that cycle would beat a Carnot engine
- $-\oint \delta Q/T$ measures the irreversibility of the cycle

Testing a cycle: 1000 kJ from 800 K, sink at 300 K

Heat rejected	$\oint \delta Q/T = 1000/800 - Q_2/300$	Verdict
500 kJ	$1.25 - 1.667 = -0.417 \text{ kJ/K} < 0$	Irreversible but possible ($\eta = 50\%$ against Carnot 62.5%)
375 kJ	$1.25 - 1.25 = 0$	Reversible cycle – the Carnot limit
300 kJ	$1.25 - 1.0 = +0.25 \text{ kJ/K} > 0$	Impossible

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Entropy – a property of the system

Entropy is a property

- Go $1 \rightarrow 2$ by reversible path A, return by reversible path B: a reversible cycle, so $\oint \delta Q/T = 0$
- Return instead by reversible path C: again $\oint \delta Q/T = 0$
- Subtract: $\int_B \delta Q/T = \int_C \delta Q/T$ – **independent of the path**, so it is the change of a **property**

KEY POINT Clausius named this property entropy, S .

Definition of entropy

$$dS = (\delta Q/T)_{\text{rev}}$$

$$S_2 - S_1 = \int (\delta Q/T)_{\text{rev}} \text{ from 1 to 2}$$

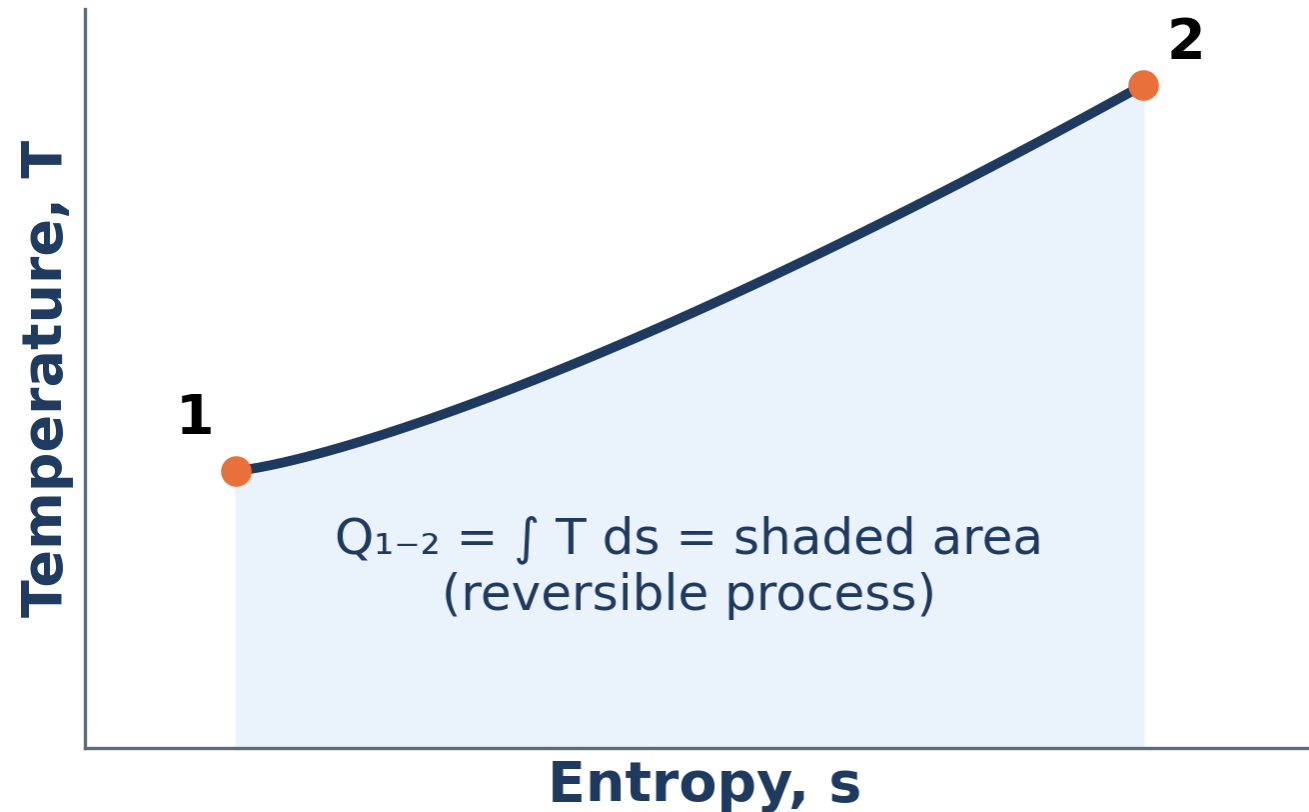
[kJ/K]

- Extensive property; specific entropy $s = S/m$ in kJ/kg K
- Only **changes** are defined; engineering uses a datum (steam: $s = 0$ for saturated liquid at the triple point)
- ΔS is the same for a reversible or an irreversible process between the same states – evaluate it along **any convenient reversible path**

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The T–s diagram

Heat transfer as an area on the T–s diagram

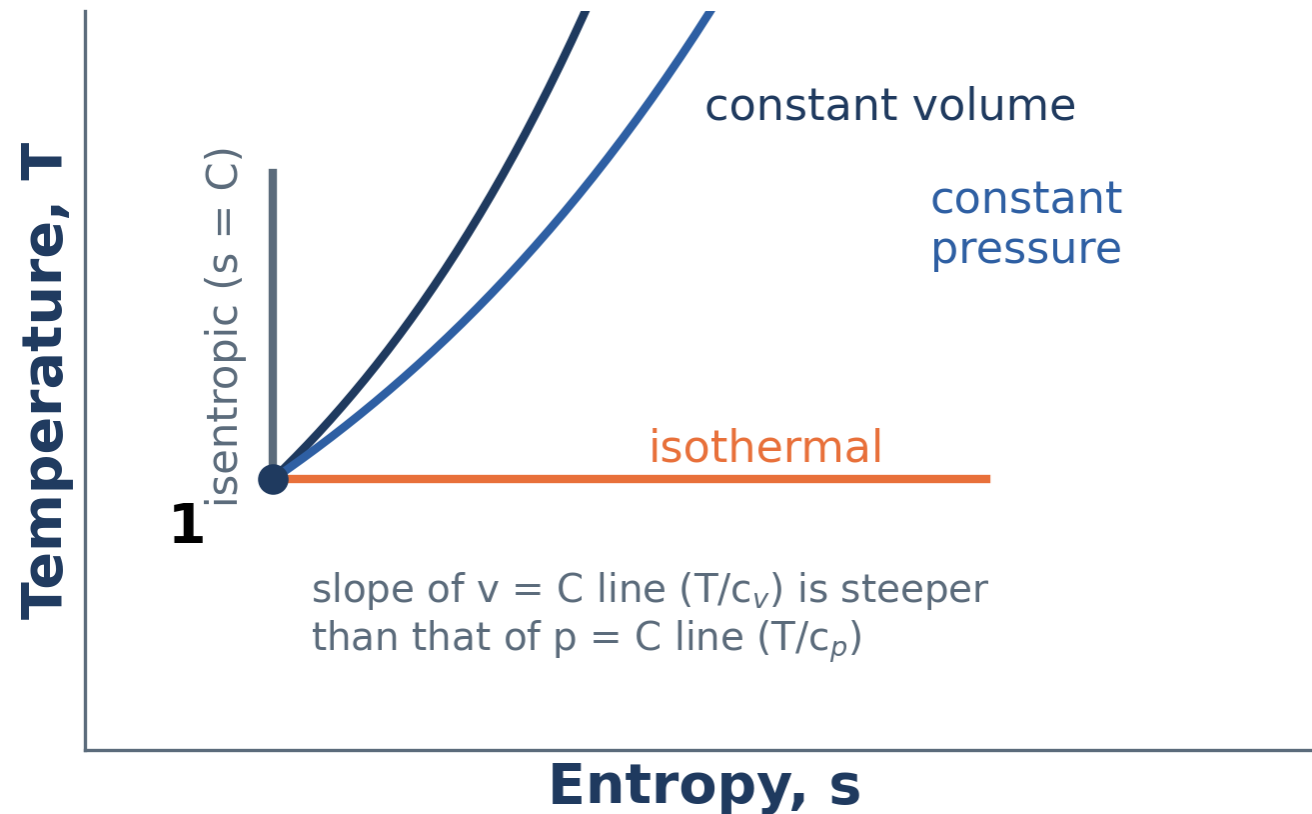


$\delta Q_{\text{rev}} = T dS$, so $Q_{1-2} = \int T dS = \text{the area under the process curve} - \text{just as work is the area under the } p\text{--}V \text{ curve}$

Reading the T–s diagram

- Entropy increases $\Rightarrow$ the system **receives** heat; decreases $\Rightarrow$ it **rejects** heat
- **Reversible adiabatic** ($\delta Q = 0$) $\Rightarrow$ constant entropy: an **isentropic** process, a vertical line
- **Reversible isothermal**: a horizontal line, $Q = T (S_2 - S_1)$
- For a cycle the enclosed area is the net heat = net work; the Carnot cycle is a rectangle

Ideal-gas processes on the T–s plane



Slope of a constant-volume line is T/c_v , of a constant-pressure line T/c_p – since $c_p > c_v$ the constant-volume line is steeper

$dS = \delta Q/T$ holds only for a reversible process

For an irreversible process $dS > \delta Q/T$: entropy is created inside the system by irreversibility, in addition to the entropy carried in by heat.

- Entropy is transferred only by **heat**, never by work – work is entropy-free energy, the deepest reason why it is the higher grade

Summary

- Reversible cycle $\Sigma Q/T = 0$; irreversible $\Sigma Q/T < 0$
- Clausius inequality: $\oint \delta Q/T \leq 0$; > 0 is impossible
- $\int \delta Q/T$ along reversible paths is path-independent $\Rightarrow$ entropy is a property
- $dS = (\delta Q/T)_{\text{rev}}$ in kJ/K; only changes matter
- $Q_{\text{rev}} = \int T dS = \text{area under the } T-s \text{ curve}$; isentropic = vertical line
- Irreversible: $dS > \delta Q/T$; entropy travels with heat, never with work

Next: Tds equations and entropy changes

- $T ds = du + p dv$ and $T ds = dh - v dp$
- Entropy change of ideal gases, solids, liquids and phase changes
- Entropy on the T–s diagram for the standard processes